

Background: chemistry and process ontology of free-radical photopolymerization

Lab note for the Photopolymerization modelling campaign
Masters-level primer (not a results paper)

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Abstract

This note is a *background* for later modelling. It does not report new experiments and it does not describe software. It states, in chemical and process-engineering language, *what objects exist* when a multifunctional acrylate resin is turned into a polymer by light, and *which of those objects must not be treated as the same thing*. The intended reader has undergraduate physical chemistry and transport, but need not have worked on stereolithography or hydrogels. The literature corpus that motivates the ontology is the set of papers collected for this campaign, especially the five local reports on first-principles kinetics [Dobson and Bowman, 2024], grayscale digital-light processing [Montgomery et al., 2022], scanning-mask projection [Meenakshisundaram et al., 2020], photopolymerization in flow [Slutzky et al., 2019], and hydrogel network mechanics [Zhu et al., 2020], together with the oxygen, reciprocity, and gelation papers those reports rest on.

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1 What we are trying to predict

A photopolymerizable liquid is a mixture that stays liquid in the dark and becomes a covalently connected solid (a gel or a densely crosslinked plastic) where and when it is illuminated, provided enough of the right chemical events occur. The engineering goal is to *place* that solid: in a vat printer, as a layer of a part; in a microfluidic channel, as a fiber or a patch; in a hydrogel, as a spatial map of stiffness.

A tempting one-number description is the *dose*

$$E = \int I(t) dt, \tag{1}$$

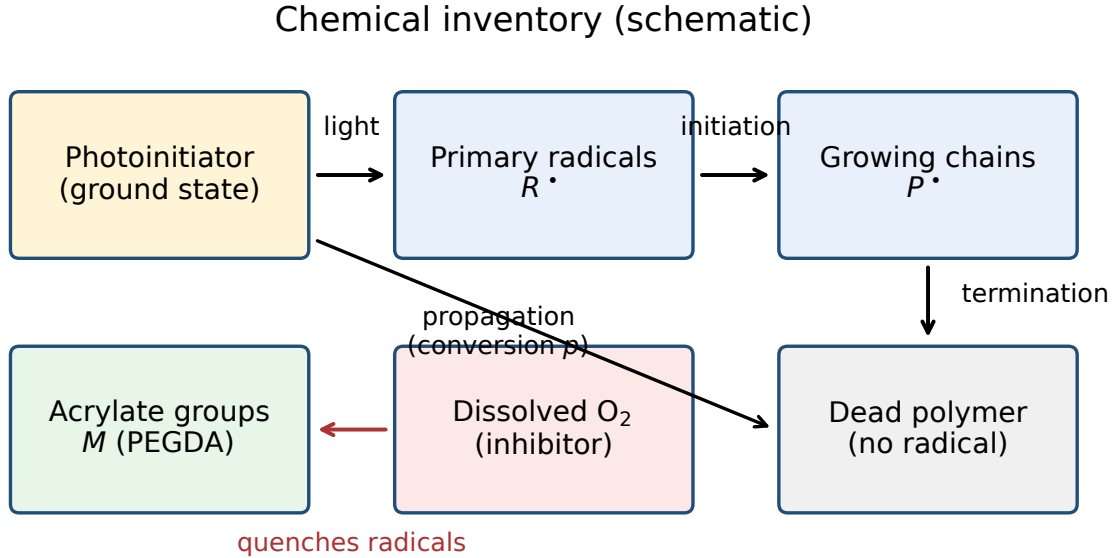
where I is the irradiance (energy, or moles of photons, arriving per area per time) at a point on the resin. Stereolithography practice has long used a threshold form of this idea: a working curve relating cure depth to $\ln(E/E_c)$ [Jacobs, 1992]. That description is useful as a machine calibration for a *given* resin, optics, and speed. It is not a material state. Two parcels of resin can receive the same E and still differ in how many acrylate groups have reacted, whether a sample-spanning network exists, and how stiff the gel is. The rest of this note explains why, by naming the chemical inventory, the competing rates, the transport of light and oxygen, and the difference between chemical conversion and a mechanically active network.

PEGDA as the running example. Poly(ethylene glycol) diacrylate (PEGDA) is a short chain with an acrylate group at each end. Each acrylate carries a carbon–carbon double bond that can add to a radical. Because there are two such groups per molecule, chain growth can

connect different PEGDA molecules into a branched, then crosslinked, network [Zhu et al., 2020, Montgomery et al., 2022, Slutzky et al., 2019]. PEGDA is *not* one resin: molecular weight (for example ≈ 250 versus $\approx 575 \text{ g mol}^{-1}$), water content, photoinitiator, and atmosphere change both the kinetics and the gel point. Numbers fitted to one formulation must not be treated as universal PEGDA constants.

2 The chemical inventory

Figure 1 is a cartoon of the species that matter for a typical free-radical acrylate photopolymerization. Light does not “glue PEGDA together” in one step. Light is absorbed by a *photoinitiator*. The initiator produces *radicals*. Radicals add to acrylate double bonds (*propagation*), which is the step that consumes monomer and builds polymer. Radicals also destroy one another (*termination*) and are scavenged by dissolved oxygen (*inhibition*). Only the competition among these processes decides how much conversion occurs for a given illumination.



Light does not polymerize PEGDA directly. It generates radicals that compete for acrylate addition versus oxygen scavenging and radical-radical termination.

Figure 1: Schematic chemical inventory for a free-radical PEGDA photopolymerization. Boxes are species classes, not a complete kinetic scheme. Conversion p counts reacted acrylate groups; it is not yet gelation or modulus.

2.1 Photoinitiators: Type I versus Type II

Photoinitiators are classified by how they generate the first radicals after absorbing a photon [Dadashi-Silab et al., 2016, Zhu et al., 2020].

Type I A single molecule undergoes bond cleavage and yields (typically two) initiating radicals. Irgacure 819 (phenylbis(2,4,6-trimethylbenzoyl)phosphine oxide), used by Montgomery et al. [2022] and in the illustrations of Dobson and Bowman [2024], is in this class. A compact description of photolysis is



with a rate proportional to the local irradiance and to the remaining initiator concentration [Montgomery et al., 2022].

Type II The absorber is excited to a triplet and must react with a co-initiator (often an amine) to produce radicals. Eosin Y with triethanolamine, the system of Zhu et al. [2020], is Type II. The excited dye can also return to the ground state or be irreversibly bleached to a leuco product. Bleaching therefore changes *both* the initiation rate *and* the optical absorption of the sample as curing proceeds.

These are different chemical objects. A rate law written for Irgacure cleavage is not interchangeable with an eosin Y bleaching cascade, even if both are used to cure PEGDA.

2.2 Monomer, conversion, and what FTIR actually measures

Let $[M]$ be the molar concentration of unreacted acrylate *functional groups* (not necessarily the concentration of PEGDA molecules). The degree of conversion is

$$p(t) = 1 - \frac{[M](t)}{[M]_0}. \quad (3)$$

In practice p is often inferred from the decay of the infrared acrylate band relative to an internal carbonyl standard [Montgomery et al., 2022]. That measurement reports *how many double bonds have disappeared*. It does not, by itself, report whether those additions connected two different chains or closed a loop on the same chain, nor whether the sample has gelled.

2.3 Radicals as a short-lived catalyst, not a stored dose

The radical concentration $[R^\bullet]$ is typically orders of magnitude smaller than $[M]$ or $[\text{O}_2]$. Radicals are generated by initiation and lost by termination and inhibition. If generation and loss nearly balance, one may write a quasi-steady estimate such as

$$[R^\bullet] \approx \sqrt{\frac{R_i}{2k_t}} \quad (4)$$

in the oxygen-free, bimolecular-termination limit, with R_i the initiation rate and k_t the termination coefficient [Zhu et al., 2020]. Equation (4) is an *approximation*, useful for insight. It fails during the first instants after the lamp is switched on, when oxygen is still abundant, and whenever k_t itself depends strongly on conversion. Keeping the radical balance as a genuine time-dependent equation is therefore a modelling choice with chemical meaning: one refuses to

assume that radicals forget their recent history [Montgomery et al., 2022, Dobson and Bowman, 2024, Slutzky et al., 2019].

2.4 Oxygen as a reagent, not a correction factor

Molecular oxygen reacts rapidly with carbon-centred radicals to give comparatively unreactive peroxy species [Decker and Jenkins, 1985, Dendukuri et al., 2008, Slutzky et al., 2019, Montgomery et al., 2022]:



Until the local oxygen inventory is depleted, propagation cannot compete. This has two engineering consequences.

First, there is an *induction period*: illumination can proceed with almost no conversion while dissolved oxygen is consumed. Second, oxygen is *replenished* by diffusion from an interface (air, an oxygen-permeable window such as PDMS, or a flowing liquid that has not yet been illuminated). Dendukuri et al. [2008] showed that in a PDMS microfluidic device the oxygen leaking through the elastomer sets a dead zone and a delay that cannot be read from a well-mixed bottle experiment. Tumbleston et al. [2015] turned a persistent oxygen-inhibited liquid layer into a manufacturing feature (continuous liquid interface production). The ontology point is the same: O_2 is a transported chemical field with sources at boundaries, not a constant “inhibition factor” multiplying the dose.

3 The kinetic skeleton (chain growth)

A minimal reaction list, following the scheme used in flow by Slutzky et al. [2019] and in grayscale printing by Montgomery et al. [2022], is:

Photolysis / initiation	$PI \xrightarrow{h\nu} R^\bullet$
Chain initiation	$R^\bullet + M \rightarrow P^\bullet$
Propagation	$P_n^\bullet + M \xrightarrow{k_p} P_{n+1}^\bullet$
Bimolecular termination	$P^\bullet + P^\bullet \xrightarrow{k_t} \text{dead polymer}$
Oxygen inhibition	$P^\bullet + O_2 \xrightarrow{k_O} \text{inactive}$

The polymerization rate (disappearance of acrylate) is then of the form

$$R_p = k_p [R^\bullet] [M]. \quad (6)$$

Everything interesting is hidden in $[R^\bullet]$ and in the coefficients k_p and k_t .

3.1 Why k_p and k_t are not constants

As conversion rises, the mixture becomes more viscous. Small monomers may still diffuse when large macroradicals can no longer find each other by centre-of-mass motion. Termination

therefore slows (*autoacceleration*: the classical Trommsdorff–Norrish gel effect), while propagation later also becomes diffusion-limited (*autodeceleration*). This picture goes back at least to analyses of diffusion-controlled termination [Benson and North, 1959] and is standard in thick-film photopolymerization models [Goodner and Bowman, 2002, Anseth et al., 1994].

At still higher conversion, radicals can “move” by adding monomer rather than by translating: *reaction-diffusion-controlled termination*, for which k_t becomes proportional to $k_p[M]$ [Anseth et al., 1994, Dobson and Bowman, 2024, Montgomery et al., 2022]. Dobson and Bowman [2024] emphasize a further refinement: short and long macroradicals do not share one k_t , because mobility depends on chain length and on whether the radical is already attached to the gel.

None of this is optional decoration if the process spans a wide conversion window. It is optional if one only asks whether a dilute flowing jet has consumed enough oxygen to gel at $\sim 2\%$ double-bond conversion [Slutzky et al., 2019]. The ontology must therefore allow *reduced* kinetics for a narrowly posed question, without pretending that the reduced coefficients are the true molecular rate constants of PEGDA in every geometry.

3.2 Thiol–ene is a different polymerization

Acrylate chain growth is not the only photocurable chemistry. Thiol–ene reactions have a step-growth character, different oxygen tolerance, and different network statistics [Cramer and Bowman, 2001]. They must not be forced into the same four-species chain-growth skeleton without rewriting both the rates and the gelation criterion. This note keeps them out of the PEGDA object.

4 Dose is not a state: the failure of exposure reciprocity

Photographic reciprocity says that the same $E = It$ produces the same blackening. If photopolymerization were a one-step photoreaction with no competing second-order losses, something similar might hold for p . Free-radical polymerization is not that reaction.

Bimolecular termination is second order in $[R^\bullet]$. Higher irradiance produces a higher radical concentration, hence a higher *fraction* of radicals lost to termination rather than to propagation. Oxygen inhibition is also first order in $[R^\bullet]$. The conversion therefore depends on the *shape* of $I(t)$, not only on its integral.

Wydra et al. [2014] tested this on a BisGMA/TEGDMA dimethacrylate. At equal dose, different irradiances produced different conversions and shrinkage stresses; the system “does not and should not be expected” to obey a reciprocity law, because bimolecular termination dominates. Related kinetic analyses include the role of oxygen, viscosity, and time-varying intensity [Lin et al., 2019]. Dobson and Bowman [2024] show a complementary symptom: the apparent scaling of R_p with I can shift from roughly $\sqrt{I}$ (classic bimolecular termination) toward a more linear I dependence when oxygen makes radical loss pseudo-first-order.

Figure 2 is a *cartoon* of that idea, not a plot of anyone’s FTIR data. The scientific statement is: if two illumination protocols are declared equivalent because $I_1 t_1 = I_2 t_2$, one has already

assumed a constitutive law that acrylates typically violate.

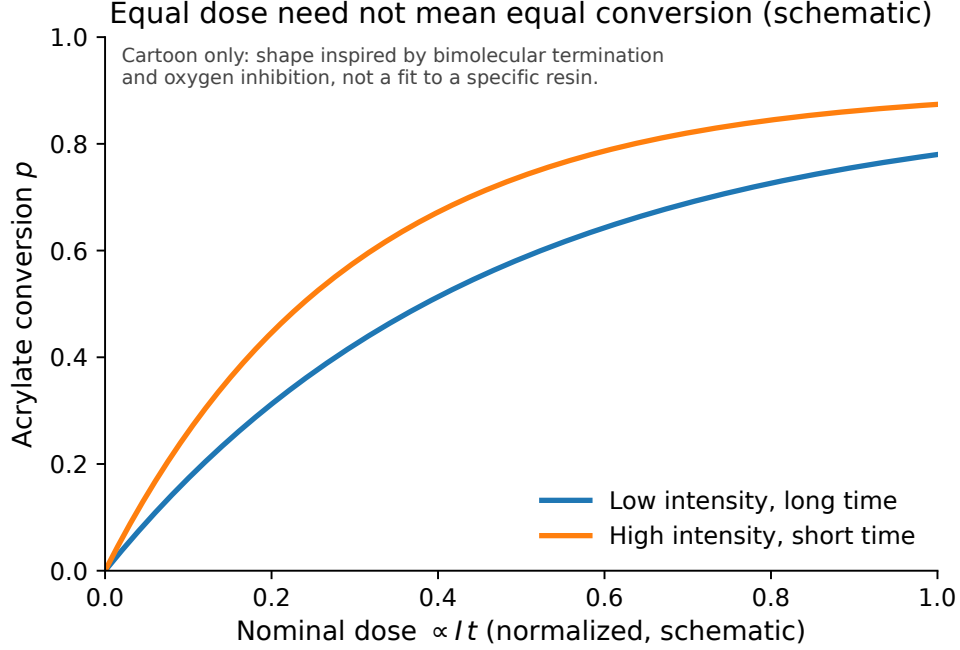


Figure 2: Schematic only. Two protocols with the same nominal dose need not yield the same conversion when termination is bimolecular and oxygen participates [Wydra et al., 2014, Lin et al., 2019, Dobson and Bowman, 2024]. The curves are not experimental measurements.

5 Light as a field in a participating medium

Irradiance is not a property of the lamp alone. It is a field $I(\mathbf{x}, t)$ inside an absorbing (and sometimes scattering) mixture.

5.1 Beer–Lambert attenuation that evolves with chemistry

If scattering and emission are negligible and the beam is treated as unidirectional, a steady radiative balance reduces to

$$\frac{\partial I}{\partial z} + A(\mathbf{x}, t) I = 0, \quad (7)$$

with z along the beam [Montgomery et al., 2022]. The attenuation A is a sum of contributions: remaining photoinitiator, unreacted monomer, polymer, and any dedicated absorber dye. As initiator is consumed, A can fall (*photobleaching*, especially for Type II dyes [Zhu et al., 2020]). As polymer forms, A can rise if the polymer absorbs more than the monomer [Montgomery et al., 2022]. In an optically thick sample the result is a polymerization *front* that advances from the illuminated face [Cabral and Douglas, 2005, Zhu et al., 2020].

Equation (7) is already an engineering reduction of radiative transfer. It is appropriate for a clear PEGDA resin. It is not appropriate for a ceramic-filled slurry, where scattering dominates.

5.2 Pixels, grayscale, and the point-spread function

A digital micromirror device (DMD) does not paint a perfectly sharp bitmap onto the resin. Each micromirror illuminates a small neighbourhood. Montgomery et al. [2022] represent that neighbourhood as a Gaussian and obtain the surface irradiance by summing all pixels, including grayscale (partial on-time of a mirror). Meenakshisundaram et al. [2020] insist that the true kernel need not be Gaussian: measured irradiance from a projection head can have a flatter core and more bleed than a first-order Gaussian, because of mirror tilt, imperfect collimation, and the relay optics [citing Zhou and Chen, 2009].

The ontology distinction:

- the *commanded bitmap* is a discrete pattern of grayscale values;
- the *surface irradiance* is a continuous field obtained by convolution with the optical kernel and, if the head or the image moves, by a time integral;
- the *in-resin irradiance* is that surface field after attenuation (7);
- the *chemical conversion* is the result of the kinetic skeleton driven by the in-resin irradiance.

Collapsing all four into a Jacobs working curve throws away the objects one needs in order to predict grayscale gradients and overcure [Montgomery et al., 2022, Meenakshisundaram et al., 2020].

6 Two independent motions

It is easy to say “the process is moving.” Two different velocities must be named (Fig. 3).

Pattern velocity $\mathbf{v}_{\text{pattern}}$. In scanning-mask projection, the projector (or the bitmap on the DMD) translates so that a large area can be addressed with a small chip [Meenakshisundaram et al., 2020]. Frame rate, scan speed, and projected pixel pitch together decide how long a resin point sees each column of the movie. One can even scroll the bitmap to *cancel* stage motion, so that a feature is stationary in the vat while the hardware moves.

Material velocity $\mathbf{u}$. In a microfluidic jet, oligomer is carried through a stationary UV spot of length L at speed U [Slutzky et al., 2019]. The residence time in the light is $t_L = L/U$. If the lamp is pulsed for a duration t_{uv} , the relevant dimensionless exposure is t_{uv}/t_L .

A material element at position $\mathbf{x}_p(t)$ with $d\mathbf{x}_p/dt = \mathbf{u}$ sees

$$I_p(t) = I(\mathbf{x}_p(t), t). \quad (8)$$

That sampled history is what the kinetic skeleton should be fed. Calling both “speed” hides whether one is moving the lamp, the mask, or the liquid.

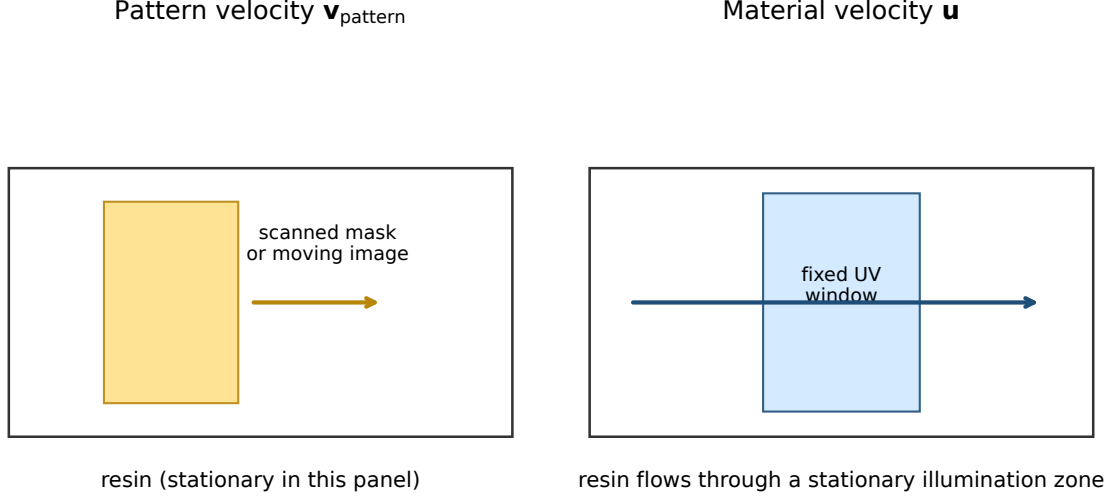


Figure 3: Left: the illumination pattern moves over a vat (scanning mask or scrolled image) while the resin is, to first approximation, still [Meenakshisundaram et al., 2020]. Right: the resin flows through a fixed light window [Slutzky et al., 2019]. A fluid element experiences the *relative* motion of pattern and material.

7 Flow replenishes oxygenated resin

In a closed, stagnant film, dissolved oxygen is a finite inventory (plus whatever diffuses from the walls). In a continuous flow, fresh, oxygenated liquid continually enters the illumination zone. Slutzky et al. [2019] make this the condition for fiber formation: polymerization remains negligible until oxygen is depleted, and the critical processing speed scales with how fast photolysis can eat the incoming O_2 inventory,

$$U_c \sim \frac{L \phi \varepsilon I [\text{PI}]_0}{[\text{O}_2]_0}, \quad (9)$$

up to the precise photophysical factors in their analysis. They also argue that, for their jet diameter and speeds, oxygen *diffusion into the jet during the residence time* is a small correction relative to advection of the dissolved inventory. That statement is geometry-specific. In a thin film against PDMS, the opposite limit can hold: diffusion from the wall dominates [Dendukuri et al., 2008].

The ontology: *advection* and *diffusion* of oxygen are different supply routes. A model that includes only one of them is answering a different experiment.

8 Conversion is not gelation is not modulus

Figure 4 is the single most important hierarchy in this note.

Do not collapse these layers into a single “dose”

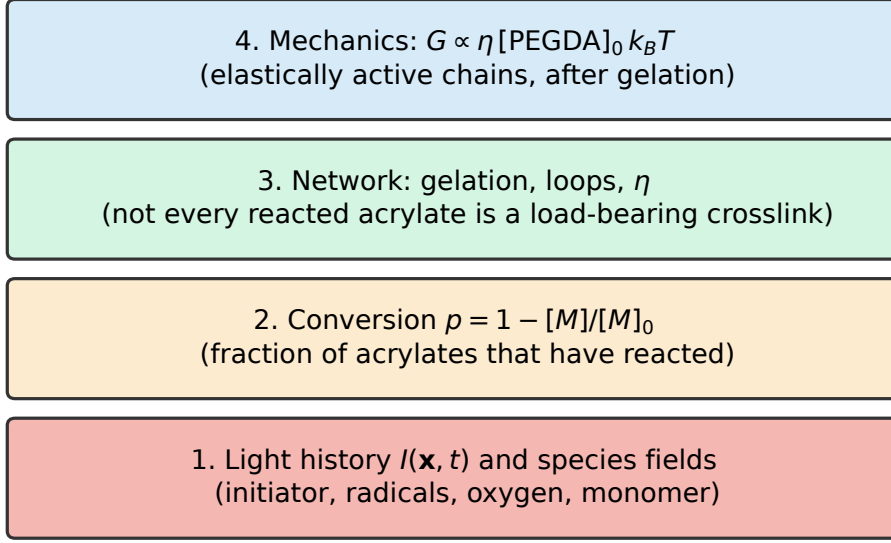


Figure 4: Four layers that must not be identified. Light history and species fields determine conversion p ; p and composition determine whether a sample-spanning network exists and how many chains are elastically active; modulus is a mechanical consequence of that network [Zhu et al., 2020, Macosko and Miller, 1976, Zhong et al., 2016].

8.1 Gelation

Gelation is the appearance of a sample-spanning covalently connected cluster: the weight-average molecular weight diverges [Flory, 1941, Macosko and Miller, 1976]. For a chain-growth polymerization of a multifunctional acrylate, that can occur at *low* conversion because each chain can connect many monomers.

Two operational definitions in the present corpus must not be mixed without comment.

- Slutzky et al. [2019] take gelation, following earlier microfluidic work, when about 2% of double bonds have reacted ($[M]/[M]_0 = 0.98$). That is a convenient kinetic threshold for “the jet no longer behaves as a liquid” under their conditions.
- Zhu et al. [2020] compute a conversion at gelation from a recursive Macosko–Miller argument that includes intramolecular loops. If θ is the probability that a reacted acrylate connected *intermolecularly* rather than as a loop, then

$$P_{\text{gel}} = \frac{1 - \theta}{2\theta}, \quad \theta = \frac{[\text{PEGDA}]_0}{[\text{PEGDA}]_0 + 2C_0}, \quad (10)$$

so $P_{\text{gel}} = C_0/[\text{PEGDA}]_0$. If the precursor is more dilute than C_0 , $P_{\text{gel}} \geq 1$: the sample never gels even at full acrylate conversion, because too many additions are wasted as loops.

Both can be legitimate. They are not the same predicate.

8.2 Loops and elastically active density

Even after gelation, not every reacted acrylate contributes a load-bearing chain. An intramolecular loop links a strand to itself and does not transmit stress the way an intermolecular crosslink does [Zhong et al., 2016, Zhu et al., 2020]. Zhu et al. [2020] construct a fractional crosslinking density η from the probabilities that arms lead to the infinite network, and then write a phantom-network shear modulus

$$G = \eta N_A [\text{PEGDA}]_0 k_B T. \quad (11)$$

The ideal affine or phantom network without loops is an upper bound; real PEGDA gels sit below it. Fitting G as a smooth function of dose, skipping η , hides the concentration-dependent gel point that Zhu et al. [2020] used to design stiffness gradients.

8.3 Heat, shrinkage, and swelling

Polymerization is exothermic; thick or intensely illuminated samples heat, and rates depend on temperature [Goodner and Bowman, 2002, Dobson and Bowman, 2024]. Conversion also shrinks the volume and, in hydrogels, the network later imbibes solvent. Those are real couplings. They are *downstream* of a trustworthy conversion field. Treating them as the first objects to predict, before $p(\mathbf{x}, t)$ is even defined, inverts the hierarchy of Fig. 4.

9 How the five campaign papers partition the ontology

The local literature folder is not five versions of one model. Each paper owns a different layer.

A predictive description of a *flowing grayscale PEGDA hydrogel* needs all four chemical–physical layers—kinetics, optics, transport, network—but it should inherit each layer from the paper that actually resolved it, not average their rate constants.

10 What must not be identified

For a masters-level reader, the practical discipline is a list of forbidden equivalences.

1. $E = \int I dt \not\equiv$ conversion p . Reciprocity fails when termination is bimolecular and oxygen participates [Wydra et al., 2014, Lin et al., 2019].
2. Commanded bitmap $\not\equiv$ surface irradiance $\not\equiv$ in-resin $I(\mathbf{x}, t)$ [Meenakshisundaram et al., 2020, Montgomery et al., 2022].
3. Pattern velocity $\not\equiv$ resin velocity [Meenakshisundaram et al., 2020, Slutzky et al., 2019].

Table 1: Division of labour among the five unique reports in the campaign corpus. “Owns” means: this is the layer the paper actually closes with equations and experiments. It does not mean the other layers are false.

Report	Layer owned
Dobson and Bowman [2024]	High-resolution kinetic ontology: many species, free-volume transport, short versus long radicals, oxygen, heat, one- and three-dimensional balances. Chemistry illustrated on HDDA, not PEGDA.
Montgomery et al. [2022]	Spatial grayscale printing: four tracked species, conversion-dependent k_p and k_t , evolving Beer–Lambert field, pixel overlap, lateral diffusion that blurs commanded edges. PEGDA 250 / Irgacure 819.
Meenakshisundaram et al. [2020]	Projection kinematics: measured (possibly non-Gaussian) pixel irradiance, scanning mask, discrete frames, dose accumulation. Cure profile compared to printed dimensions; chemistry is not the species skeleton above.
Slutzky et al. [2019]	Flowing material through a fixed light window: advection of initiator, radicals, monomer, and oxygen; critical speed; pulse duration versus residence time. PEGDA 575 jet; oxygen diffusion into the jet argued small.
Zhu et al. [2020]	Type II eosin Y bleaching plus conversion-to-gelation-to-modulus closure with loops. Frontal photopolymerization in depth. PEGDA 575 hydrogel; no advection.

4. Dissolved oxygen inventory $\neq$ oxygen leaking through a wall [Slutzky et al., 2019, Dendukuri et al., 2008].
5. Type I photolysis $\neq$ Type II dye/amine bleaching [Dadashi-Silab et al., 2016, Zhu et al., 2020, Montgomery et al., 2022].
6. PEGDA 250 in a dry thin film $\neq$ PEGDA 575 in water $\neq$ HDDA bulk [Montgomery et al., 2022, Zhu et al., 2020, Dobson and Bowman, 2024].
7. Acrylate conversion $p \neq$ gelation $\neq$ elastically active density $\eta \neq$ modulus G [Zhu et al., 2020, Flory, 1941, Zhong et al., 2016].
8. A 2% double-bond threshold [Slutzky et al., 2019] $\neq$ the Macosko–Miller $P_{\text{gel}}([\text{PEGDA}]_0)$ [Zhu et al., 2020].
9. Chain-growth acrylate kinetics $\neq$ thiol–ene step growth [Cramer and Bowman, 2001].
10. Jacobs working-curve depth [Jacobs, 1992] $\neq$ a species-resolved polymer state.

11 Implications for a predictive description

The ontology above already implies an order of questions, still without reference to implementation.

1. **Local constitutive response.** For a prescribed $I(t)$, how do initiator, radicals, oxygen, and acrylate evolve? This is the question Dobson and Bowman [2024] answer at high

chemical resolution and Montgomery et al. [2022] answer with four species and conversion-dependent coefficients. Until this map is trusted—including an equal-dose test that is *allowed to fail*—spatial complexity only conceals error.

2. **Optical boundary condition.** What irradiance actually sits on $z = 0$ as a function of bitmap, pixel kernel, and pattern motion [Meenakshisundaram et al., 2020]? Then, how does $A(p, [\text{PI}])$ carry that field into the depth [Montgomery et al., 2022, Zhu et al., 2020]?
3. **Transport.** Which species move relative to the material? Oxygen almost always does; monomer may; network-bound radicals do not [Dobson and Bowman, 2024, Montgomery et al., 2022]. Is the supply advective [Slutzky et al., 2019] or diffusive from a wall [Dendukuri et al., 2008]?
4. **Network closure.** Given p and composition, has the sample gelled, and what is η and G [Zhu et al., 2020]? This step must not be used to patch kinetic coefficients.

The campaign choice that follows from this background is conservative: treat PEGDA chain-growth as the first chemical object; treat Dobson’s full species set as a *reference* against which reduced $k_p(p)$ and $k_t(p)$ can be judged, not as the first process model; treat scanning projection and plug flow as two different kinematic experiments; treat Zhu’s loop-aware modulus as a post-processor on conversion. Those are scientific restrictions, not software preferences.

Evidence status of this document

This note is a reading synthesis (**proposed** ontology). It does not validate any numerical implementation. Figures 1–4 are teaching cartoons. Quantitative claims about a specific PEGDA formulation remain **still required** until they are taken from a named experiment or from a paper figure under the same recipe.

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